

Runtime Results

scale	m	n	unknowns	nodes	ppn	threads	MPI ranks	total cores	N/C	matrix type	runtime seconds
small	1000	1000	1000000	1	1	1	1	1	1,000,000	AIJ	245.7
small	1000	1000	1000000	1	1	1	1	1	1,000,000		248.0
small	1000	1000	1000000	1	2	1	2	2	500,000		83.55
small	1000	1000	1000000	1	2	1	2	2	500,000		84.07
small	1000	1000	1000000	1	1	2	1	2	500,000		229.6
small	1000	1000	1000000	1	1	2	1	2	500,000		231.4
small	1000	1000	1000000	1	4	1	4	4	250,000		48.91
small	1000	1000	1000000	1	4	1	4	4	250,000		49.61
small	1000	1000	1000000	1	2	2	2	4	250,000		75.31
small	1000	1000	1000000	1	2	2	2	4	250,000		82.38
small	1000	1000	1000000	1	1	4	1	4	250,000		225.7
small	1000	1000	1000000	1	1	4	1	4	250,000		230.5
small	1000	1000	1000000	1	8	1	8	8	125,000		60.14
small	1000	1000	1000000	1	8	1	8	8	125,000		60.21
small	1000	1000	1000000	1	4	2	4	8	125,000		43.32
small	1000	1000	1000000	1	4	2	4	8	125,000		43.41
small	1000	1000	1000000	1	2	4	2	8	125,000		71.75
small	1000	1000	1000000	1	2	4	2	8	125,000		76.11
small	1000	1000	1000000	1	1	8	1	8	125,000		225.5
small	1000	1000	1000000	1	1	8	1	8	125,000		226.4
small	1000	1000	1000000	1	16	1	16	16	62,500		85.25
small	1000	1000	1000000	1	16	1	16	16	62,500		85.69
small	1000	1000	1000000	1	8	2	8	16	62,500		28.07
small	1000	1000	1000000	1	8	2	8	16	62,500		43.21
small	1000	1000	1000000	1	4	4	4	16	62,500		27.10
small	1000	1000	1000000	1	4	4	4	16	62,500		34.02
small	1000	1000	1000000	1	2	8	2	16	62,500		70.22
small	1000	1000	1000000	1	2	8	2	16	62,500		70.38
small	1000	1000	1000000	1	1	16	1	16	62,500		215.4
small	1000	1000	1000000	1	32	1	32	32	31,250		29.80

scale	m	n	unknowns	nodes	ppn	threads	MPI ranks	total cores	N/C	matrix type	runtime seconds
small	1000	1000	1000000	1	32	1	32	32	31,250		29.80
small	1000	1000	1000000	1	16	2	16	32	31,250		38.59
small	1000	1000	1000000	1	16	2	16	32	31,250		38.70
small	1000	1000	1000000	1	8	4	8	32	31,250		23.02
small	1000	1000	1000000	1	8	4	8	32	31,250		23.30
small	1000	1000	1000000	1	4	8	4	32	31,250		25.73
small	1000	1000	1000000	1	4	8	4	32	31,250		26.06
small	1000	1000	1000000	1	2	16	2	32	31,250		58.24
small	1000	1000	1000000	1	1	32	1	32	31,250		216.5
small	1000	1000	1000000	1	64	1	64	64	15,625		8.107
small	1000	1000	1000000	1	64	1	64	64	15,625		8.169
small	1000	1000	1000000	1	32	2	32	64	15,625		9.208
small	1000	1000	1000000	1	32	2	32	64	15,625		9.312
small	1000	1000	1000000	1	16	4	16	64	15,625		16.92
small	1000	1000	1000000	1	16	4	16	64	15,625		17.53
small	1000	1000	1000000	1	8	8	8	64	15,625		16.15
small	1000	1000	1000000	1	8	8	8	64	15,625		16.55
small	1000	1000	1000000	1	4	16	4	64	15,625		20.76
small	1000	1000	1000000	1	2	32	2	64	15,625		54.11
small	1000	1000	1000000	1	1	64	1	64	15,625		195.6
small	1000	1000	1000000	1	128	1	128	128	7,812		3.844
small	1000	1000	1000000	1	128	1	128	128	7,812		3.866
small	1000	1000	1000000	1	64	2	64	128	7,812		4.369
small	1000	1000	1000000	1	64	2	64	128	7,812		4.375
small	1000	1000	1000000	1	32	4	32	128	7,812		6.394
small	1000	1000	1000000	1	32	4	32	128	7,812		6.402
small	1000	1000	1000000	1	16	8	16	128	7,812		15.44
small	1000	1000	1000000	1	16	8	16	128	7,812		15.48
small	1000	1000	1000000	1	8	16	8	128	7,812		14.99
small	1000	1000	1000000	1	4	32	4	128	7,812		20.45
small	1000	1000	1000000	1	2	64	2	128	7,812		53.57
small	1000	1000	1000000	1	1	128	1	128	7,812		208.4

scale	m	n	unknowns	nodes	ppn	threads	MPI ranks	total cores	N/C	matrix type	runtime seconds
medium-small	2240	2240	5017600	1	1	1	1	1	5,017,600		1,874
medium-small	2240	2240	5017600	1	2	1	2	2	2,508,800		1,136
medium-small	2240	2240	5017600	1	1	2	1	2	2,508,800		1,742
medium-small	2240	2240	5017600	1	4	1	4	4	1,254,400		1,052
medium-small	2240	2240	5017600	1	2	2	2	4	1,254,400		1,135
medium-small	2240	2240	5017600	1	1	4	1	4	1,254,400		1,747
medium-small	2240	2240	5017600	1	8	1	8	8	627,200		1,067
medium-small	2240	2240	5017600	1	4	2	4	8	627,200		1,053
medium-small	2240	2240	5017600	1	2	4	2	8	627,200		1,148
medium-small	2240	2240	5017600	1	1	8	1	8	627,200		1,767
medium-small	2240	2240	5017600	1	16	1	16	16	313,600		810.1
medium-small	2240	2240	5017600	1	8	2	8	16	313,600		817.3
medium-small	2240	2240	5017600	1	4	4	4	16	313,600		828.8
medium-small	2240	2240	5017600	1	2	8	2	16	313,600		1,084
medium-small	2240	2240	5017600	1	32	1	32	32	156,800		330.3
medium-small	2240	2240	5017600	1	16	2	16	32	156,800		332.4
medium-small	2240	2240	5017600	1	8	4	8	32	156,800		365.3
medium-small	2240	2240	5017600	1	4	8	4	32	156,800		531.9
medium-small	2240	2240	5017600	1	64	1	64	64	78,400		135.3

scale	m	n	unknowns	nodes	ppn	threads	MPI ranks	total cores	N/C	matrix type	runtime seconds
medium-small	2240	2240	5017600	1	32	2	32	64	78,400		135.4
medium-small	2240	2240	5017600	1	16	4	16	64	78,400		156.3
medium-small	2240	2240	5017600	1	8	8	8	64	78,400		242.7
medium-small	2240	2240	5017600	1	128	1	128	128	39,200		64.02
medium-small	2240	2240	5017600	1	64	2	64	128	39,200		64.76
medium-small	2240	2240	5017600	1	32	4	32	128	39,200		76.01
medium-small	2240	2240	5017600	1	16	8	16	128	39,200		118.9
medium	3160	3160	9985600	1	1	1	1	1	9,985,600		3,690
medium	3160	3160	9985600	1	1	1	1	1	9,985,600		3,729
medium	3160	3160	9985600	1	2	1	2	2	4,992,800		2,221
medium	3160	3160	9985600	1	2	1	2	2	4,992,800		2,253
medium	3160	3160	9985600	1	1	2	1	2	4,992,800		3,461
medium	3160	3160	9985600	1	1	2	1	2	4,992,800		3,463
medium	3160	3160	9985600	1	4	1	4	4	2,496,400		2,064
medium	3160	3160	9985600	1	2	2	2	4	2,496,400		2,235
medium	3160	3160	9985600	1	2	2	2	4	2,496,400		2,252
medium	3160	3160	9985600	1	1	4	1	4	2,496,400		3,473
medium	3160	3160	9985600	1	1	4	1	4	2,496,400		3,481
medium	3160	3160	9985600	1	8	1	8	8	1,248,200		2,145
medium	3160	3160	9985600	1	4	2	4	8	1,248,200		2,068
medium	3160	3160	9985600	1	2	4	2	8	1,248,200		2,273
medium	3160	3160	9985600	1	2	4	2	8	1,248,200		2,277
medium	3160	3160	9985600	1	1	8	1	8	1,248,200		3,410
medium	3160	3160	9985600	1	1	8	1	8	1,248,200		3,522
medium	3160	3160	9985600	1	16	1	16	16	624,100		1,838
medium	3160	3160	9985600	1	8	2	8	16	624,100		1,797
medium	3160	3160	9985600	1	4	4	4	16	624,100		1,763
medium	3160	3160	9985600	1	2	8	2	16	624,100		2,155

scale	m	n	unknowns	nodes	ppn	threads	MPI ranks	total cores	N/C	matrix type	runtime seconds
medium	3160	3160	9985600	1	32	1	32	32	312,050		804.0
medium	3160	3160	9985600	1	16	2	16	32	312,050		809.7
medium	3160	3160	9985600	1	8	4	8	32	312,050		824.1
medium	3160	3160	9985600	1	4	8	4	32	312,050		1,090
medium	3160	3160	9985600	1	64	1	64	64	156,025		329.7
medium	3160	3160	9985600	1	32	2	32	64	156,025		331.9
medium	3160	3160	9985600	1	16	4	16	64	156,025		366.2
medium	3160	3160	9985600	1	8	8	8	64	156,025		530.8
medium	3160	3160	9985600	1	128	1	128	128	78,012		136.5
medium	3160	3160	9985600	1	64	2	64	128	78,012		136.7
medium	3160	3160	9985600	1	32	4	32	128	78,012		157.4
medium	3160	3160	9985600	1	16	8	16	128	78,012		243.1
large	4500	4500	20250000	1	1	1	1	1	20,250,000		7,537
large	4500	4500	20250000	1	1	1	1	1	20,250,000		7,555
large	4500	4500	20250000	1	2	1	2	2	10,125,000		4,516
large	4500	4500	20250000	1	2	1	2	2	10,125,000		4,525
large	4500	4500	20250000	1	1	2	1	2	10,125,000		7,018
large	4500	4500	20250000	1	1	2	1	2	10,125,000		7,044
large	4500	4500	20250000	1	4	1	4	4	5,062,500		4,191
large	4500	4500	20250000	1	4	1	4	4	5,062,500		4,192
large	4500	4500	20250000	1	2	2	2	4	5,062,500		4,569
large	4500	4500	20250000	1	2	2	2	4	5,062,500		4,570
large	4500	4500	20250000	1	1	4	1	4	5,062,500		7,074
large	4500	4500	20250000	1	1	4	1	4	5,062,500		7,102
large	4500	4500	20250000	1	8	1	8	8	2,531,250		4,288
large	4500	4500	20250000	1	8	1	8	8	2,531,250		4,290
large	4500	4500	20250000	1	4	2	4	8	2,531,250		3,550
large	4500	4500	20250000	1	4	2	4	8	2,531,250		4,202
large	4500	4500	20250000	1	2	4	2	8	2,531,250		4,324
large	4500	4500	20250000	1	2	4	2	8	2,531,250		4,625
large	4500	4500	20250000	1	1	8	1	8	2,531,250		7,144
large	4500	4500	20250000	1	1	8	1	8	2,531,250		7,154
large	4500	4500	20250000	1	16	1	16	16	1,265,625		3,767

scale	m	n	unknowns	nodes	ppn	threads	MPI ranks	total cores	N/C	matrix type	runtime seconds
large	4500	4500	20250000	1	16	1	16	16	1,265,625		3,768
large	4500	4500	20250000	1	8	2	8	16	1,265,625		3,623
large	4500	4500	20250000	1	8	2	8	16	1,265,625		3,625
large	4500	4500	20250000	1	4	4	4	16	1,265,625		2,307
large	4500	4500	20250000	1	4	4	4	16	1,265,625		2,319
large	4500	4500	20250000	1	2	8	2	16	1,265,625		4,384
large	4500	4500	20250000	1	2	8	2	16	1,265,625		4,396
large	4500	4500	20250000	1	32	1	32	32	632,812		1,868
large	4500	4500	20250000	1	32	1	32	32	632,812		1,869
large	4500	4500	20250000	1	16	2	16	32	632,812		1,826
large	4500	4500	20250000	1	16	2	16	32	632,812		1,827
large	4500	4500	20250000	1	8	4	8	32	632,812		1,793
large	4500	4500	20250000	1	8	4	8	32	632,812		1,794
large	4500	4500	20250000	1	4	8	4	32	632,812		2,209
large	4500	4500	20250000	1	4	8	4	32	632,812		2,220
large	4500	4500	20250000	1	64	1	64	64	316,406		825.7
large	4500	4500	20250000	1	64	1	64	64	316,406		826.7
large	4500	4500	20250000	1	32	2	32	64	316,406		829.9
large	4500	4500	20250000	1	32	2	32	64	316,406		831.1
large	4500	4500	20250000	1	16	4	16	64	316,406		846.0
large	4500	4500	20250000	1	16	4	16	64	316,406		848.1
large	4500	4500	20250000	1	8	8	8	64	316,406		1,115
large	4500	4500	20250000	1	8	8	8	64	316,406		1,117
large	4500	4500	20250000	1	128	1	128	128	158,203		340.1
large	4500	4500	20250000	1	128	1	128	128	158,203		340.8
large	4500	4500	20250000	1	64	2	64	128	158,203		341.6
large	4500	4500	20250000	1	64	2	64	128	158,203		342.1
large	4500	4500	20250000	1	32	4	32	128	158,203		376.0
large	4500	4500	20250000	1	32	4	32	128	158,203		376.6
large	4500	4500	20250000	1	16	8	16	128	158,203		547.0
large	4500	4500	20250000	1	16	8	16	128	158,203		547.2
very- large	6400	6400	40960000	1	1	1	1	1	40,960,000		15,410

scale	m	n	unknowns	nodes	ppn	threads	MPI ranks	total cores	N/C	matrix type	runtime seconds
very-large	6400	6400	40960000	1	1	1	1	1	40,960,000		15,420
very-large	6400	6400	40960000	1	2	1	2	2	20,480,000		9,131
very-large	6400	6400	40960000	1	2	1	2	2	20,480,000		9,147
very-large	6400	6400	40960000	1	1	2	1	2	20,480,000		14,150
very-large	6400	6400	40960000	1	1	2	1	2	20,480,000		14,190
very-large	6400	6400	40960000	1	4	1	4	4	10,240,000		8,464
very-large	6400	6400	40960000	1	2	2	2	4	10,240,000		9,219
very-large	6400	6400	40960000	1	2	2	2	4	10,240,000		9,272
very-large	6400	6400	40960000	1	1	4	1	4	10,240,000		14,250
very-large	6400	6400	40960000	1	1	4	1	4	10,240,000		14,280
very-large	6400	6400	40960000	1	8	1	8	8	5,120,000		8,702
very-large	6400	6400	40960000	1	4	2	4	8	5,120,000		8,503
very-large	6400	6400	40960000	1	2	4	2	8	5,120,000		9,344
very-large	6400	6400	40960000	1	2	4	2	8	5,120,000		9,376
very-large	6400	6400	40960000	1	1	8	1	8	5,120,000		13,840
very-large	6400	6400	40960000	1	1	8	1	8	5,120,000		14,410
very-large	6400	6400	40960000	1	16	1	16	16	2,560,000		7,558
very-large	6400	6400	40960000	1	8	2	8	16	2,560,000		7,375
very-large	6400	6400	40960000	1	4	4	4	16	2,560,000		7,311

scale	m	n	unknowns	nodes	ppn	threads	MPI ranks	total cores	N/C	matrix type	runtime seconds
very-large	6400	6400	40960000	1	2	8	2	16	2,560,000		8,845
very-large	6400	6400	40960000	1	32	1	32	32	1,280,000		3,825
very-large	6400	6400	40960000	1	16	2	16	32	1,280,000		3,685
very-large	6400	6400	40960000	1	8	4	8	32	1,280,000		3,651
very-large	6400	6400	40960000	1	4	8	4	32	1,280,000		4,484
very-large	6400	6400	40960000	1	64	1	64	64	640,000		1,905
very-large	6400	6400	40960000	1	32	2	32	64	640,000		1,859
very-large	6400	6400	40960000	1	16	4	16	64	640,000		1,826
very-large	6400	6400	40960000	1	8	8	8	64	640,000		2,271
very-large	6400	6400	40960000	1	128	1	128	128	320,000		846.9
very-large	6400	6400	40960000	1	64	2	64	128	320,000		850.8
very-large	6400	6400	40960000	1	32	4	32	128	320,000		866.4
very-large	6400	6400	40960000	1	16	8	16	128	320,000		1,140